

Ajay Patel

(973) 617-7032 | ajpat8703@gmail.com | [Portfolio](#) | [LinkedIn](#) | [GitHub](#)

EDUCATION

University of California, Berkeley

Master of Information and Data Science – GPA: 4.00/4.00

January 2025 – August 2026

University of Rochester

Bachelor of Science, Data Science Major – GPA: 3.99/4.00

August 2021 – December 2024

EXPERIENCE

Pittsburgh Pirates

Junior Data Scientist

March 2025 – Present

- Developed and deployed PyTorch CNN models that estimate catch and infield out probabilities from player-tracking and fielder-relative spatial features. Integrated outputs into internal applications and SQL tables used for defensive evaluation.
- Built and deployed a PyTorch-based positioning model for the 2026 season that recommends starting locations for all seven fielders across matchups, ballparks, and game states, jointly optimizing the defense to minimize expected run value.

Los Angeles Dodgers

Baseball Strategy and Information Intern

May 2024 – August 2024

- Developed sliding metrics in Python/SQL from tracking data to enhance game reports and player evaluation.
- Trained an XGBoost model in R to project WAR for amateur players based on signing bonus, age, and position. Analyzed residuals to identify teams that outperformed development expectations.

Optum

Technological Development Program Intern

June 2023 – August 2023

- Designed a React & Node.js web app to host behavioral therapy data, improving data accessibility for therapists
- Developed backend SQL integrations and API endpoints. Enhanced front-end using React, HTML, and CSS.

PROJECTS

PortalPoint - UC Berkeley MIDS Capstone

Data Science Lead

June 2026 – August 2026

- Led the roadmap and data science strategy for a transfer portal scouting tool, evaluating 2,500+ NCAA men's basketball players, translating domain knowledge into scheme, roster-gap, role, and team-impact fit scores.
- Developed player and team archetypes, talent and playing-time projection models, and a two-stage recommendation engine that generated more than 450,000 explainable player-program predictions.
- Collaborated with a four-person team to deploy a React, FastAPI, and PostgreSQL application using Docker and AWS CloudFront, ECS Fargate, RDS, and S3.

NFL Big Data Bowl - [Runner-Up](#) (2024), [Finalist](#) (2023)

2022 – 2024

- Developed an XGBoost-based expected pass-rush metric from pre-snap player tracking data to analyze defensive schemes. Presented work at the NFL Combine and it's now used on AWS-powered NFL broadcasts.
- Trained a PyTorch CNN to predict yards after the catch from spatial relationships across all players. Outputs were translated into opponent scouting reports and team-level visualizations.

University of Rochester Independent Studies

Research Assistant

August 2023 – December 2024

- Analyzed political bias in AI image generation tools using Python, computer vision (DeepFace), and web scraping techniques.

SKILLS

- Languages & Machine Learning: Python, R, SQL | PyTorch, XGBoost, scikit-learn, GLMs, clustering
- Application: FastAPI, Docker, AWS, Git, Airflow, MLflow, ggplot2, Matplotlib, React